

altairsim — CP/M and FDOS on an iCOM FD3712/FD3812 8" floppy

2026-08-17 · 2b99dca

```
altairsim cpm22.toml

48K CP/M 2.2 v1.0
for iCOM FD3712 and Altair

A>DIR
```

Ready-made machines for the **iCOM FD3712/FD3812 8" floppy controller** — CP/M in single and double density, and both revisions of iCOM's own FDOS disk operating system. Each boots the moment you start it (`RUN F000` for CP/M, `RUN C000` for FDOS is the machine file's whole startup, because on a real machine that was EXAMINE the PROM address and RUN).

The iCOM is **not a bit-shifting floppy card** like the Tarbell or VersaFloppy. It is a **command/handshake** controller: it buffers a whole sector, and the CPU moves bytes through two ports (`C0h` – `C1h`) while the operating system's disk driver runs from a **boot PROM** up in high memory (`F000` for CP/M, `C000` for FDOS). The single-density **FD3712** disk is $77 \times 26 \times 128 = 256,256$ bytes; the double-density **FD3812** disk is mixed density — a single-density track 0 then double-density tracks 1–76 — 509,184 bytes. See [docs/boards/icom-fd3712.md](#) .

`^E` (ATTN) takes the keyboard back to the monitor at any point; `RUN` resumes. `^C` belongs to CP/M (it is warm boot) and CP/M gets it. FDOS's prompt is `!` . The two FDOS revisions do not share a command language: **FDOS-III** takes word commands (`LIST` for a directory), while the original **FDOS-I** takes single-letter directives — `L` lists the directory, `A` assembles, `P` prints.

The files

File	What it is
<code>cpm22.toml</code>	FD3712 single-density CP/M 2.2 . <code>base = "icom"</code> plus the disk in drive 0.
<code>cpm22-3812.toml</code>	FD3812 double-density CP/M 2.23 (Lifeboat). Swaps in the FD3812 boot PROM and the DD disk.
<code>fdos-iii.toml</code>	iCOM's own FDOS-III disk operating system. Uses the FDOS boot PROM at <code>C000</code> and boots to the <code>!</code> prompt.
<code>fdos-i.toml</code>	iCOM FDOS-I , the original revision. Same <code>C000</code> PROM as FDOS-III, but adds a high-RAM board (<code>C500-FFFF</code>) because FDOS-I loads its resident into high memory. Boots to <code>!</code> ; use single-letter directives (<code>L</code> for the directory).
<code>CPM22v1.0-3712-48K.DSK</code>	The single-density CP/M system disk.
<code>CPM22-3812-48K.dsk</code>	The double-density CP/M system disk.
<code>FDOS-III (2SI0).DSK</code>	The FDOS-III system diskette.
<code>FDOS-I (2SI0).DSK</code>	The FDOS-I system diskette.

There is no undo. Drive 0 is mounted read/write because that is what a real machine is, and the guest writes to the disk for anything you create. In a clone `git checkout` puts the image back; in the package you were handed, nothing does. Copy it first if you are about to test writes in anger, or add `readonly = true` to the drive.

The boot PROMs are Mike Douglas disassemblies and the disk images are from deramp.com (iCOM Floppy Systems). Read and write of existing disks work; like the other controllers here, this one will not lay down a **blank** format from nothing — all the shipped disks are pre-formatted.